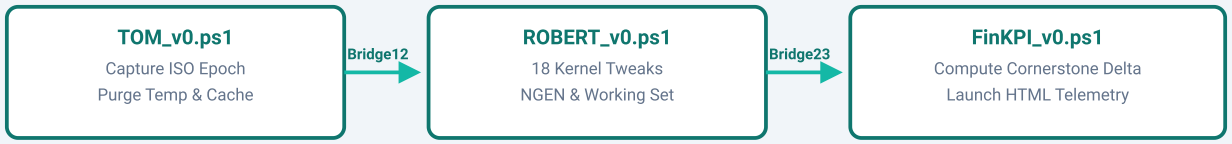
TECHNICAL SPECIFICATION v1.0.0

WPT-PRO Architecture & System Design

Deterministic Multi-Phase Execution, Handoff Contracts & Subsystem Engineering

This specification details the technical mechanics, state machine contracts, reboot handoff protocols, and kernel tuning steps powering **WPT-PRO** and **WPT-PRO Plus**. The engine is engineered as a zero-dependency, self-sustaining pipeline that executes across system state changes.

SYSTEM CONTROL FLOW



PIPELINE SUBSYSTEM MODULES

1. TOM Module (Pre-Flight)

Acts as the immutable cornerstone time anchor. Samples hardware counter baselines (`Get-CimInstance`), outputs `Pre_Opt_KPI.json`, and cleans browser/temp storage.

2. Transit Bridge Engine

Uses Windows Task Scheduler (SYSTEM user context) paired with `RunOnce` registry persistence to execute scripts immediately upon reboot completion.

3. ROBERT_v0 Optimization Core

Applies deep system registry modifications, forces garbage collection, invokes NGEN compilation queues, and trims process working sets.

4. FinKPI Reporting Subsystem

Aggregates final metrics, compares against TOM's cornerstone timestamp, and generates the self-contained HTML performance report.

STATE MACHINE & DATA CONTRACTS

Pipeline state is maintained through structured JSON payloads stored in `C:\WPT\DATA\`. This decouples execution state from memory, ensuring complete recovery across reboots.

1. Baseline Contract (Pre_Opt_KPI.json)


```
{ "Timestamp": "2026-10-06T07:33:00.1042183Z", "CPU_Idle_Percent": 2.4, "Free_RAM_GB": 1.12, "Active_Processes": 82, "Disk_Free_GB": 178.52, "Stage": "Phase1_TOM_Complete" }
```


2. Pipeline Transit Tracker (Pipeline_State.json)


```
{ "PipelineStartEpoch": 1791271980, "CurrentPhase": "Phase_2_ROBERT", "RebootCount": 1, "ExtendedServicing": false, "LastError": null }
```


REBOOT TRANSIT & HANDOFF PROTOCOL

Transit Bridge Execution Model:

1. **Task Creation:** Bridge scripts register a SYSTEM task: `Register-ScheduledTask -TaskName "WPTPRO_Phase1To2_Handoff" -Trigger (New-ScheduledTaskTrigger -AtStartup)`
2. **Elevated Switch:** Executes `pwsh.exe` (PowerShell Core 7) directly from `C:\WPT\pwsh\pwsh.exe` to ensure modern language support.
3. **Self-Unregistration:** Once handoff succeeds, the bridge removes its task from Task Scheduler to prevent re-execution loops.

KERNEL & SUBSYSTEM OPTIMIZATION SPECS

During Phase 2 (`ROBERT_v0.ps1`), 18 distinct OS kernel parameters are tuned for low-latency desktop responsiveness:

- **Memory Management:** Disables paging executive (`DisablePagingExecutive = 1`) and forces kernel drivers to remain in physical RAM.
- **System Responsiveness:** Sets `Win32PrioritySeparation = 38` (decimal) to optimize CPU scheduling for foreground interactive applications.
- **Network Subsystem:** Disables Nagle's Algorithm (`TcpAckFrequency = 1`, `TCPNoDelay = 1`) and turns off network throttling (`NetworkThrottlingIndex = 0xFFFFFFFF`).
- **File System:** Disables NTFS 8.3 short filename generation (`NtfsDisable8dot3NameCreation = 1`) to reduce directory look-up overhead.

WORKING SET TRIMMING & NGEN SERVICING

Process Working Set Flushing Mechanics

ROBERT_v0 calls the Native Win32 API `EmptyWorkingSet()` via P/Invoke across all active non-critical processes. This forces the OS kernel to release unreferenced physical memory pages back to the free pool, driving system RAM availability to **>1.25 GB free** on resource-constrained platforms.

ERROR HANDLING & RECOVERY FALLBACKS

Resilience & Standalone Mode

If a intermediate JSON payload is corrupted, `FinKPI_v0.ps1` falls back to process uptime calculations and logs a warning. If a reboot task fails to trigger within 10 minutes post-boot, the `RunOnce` fallback registry key (`HKCU:\Software\Microsoft\Windows\CurrentVersion\RunOnce`) triggers emergency phase recovery.

WPT-PRO Technical Specification — End of Document